



A Systematic Review on Data Scarcity Problem in Deep Learning: Solution and Applications

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Recent advancements in deep learning architecture have increased its utility in real-life applications. Deep learning models require a large amount of data to train the model. In many application domains, there is a limited set of data available for training neural networks as collecting new data is either not feasible or requires more resources such as in marketing, computer vision, and medical science. These models require a large amount of data to avoid the problem of overfitting. One of the data space solutions to the problem of limited data is data augmentation. The purpose of this study focuses on various data augmentation techniques that can be used to further improve the accuracy of a neural network. This saves the cost and time consumption required to collect new data for the training of deep neural networks by augmenting available data. This also regularizes the model and improves its capability of generalization. The need for large datasets in different fields such as computer vision, natural language processing, security, and healthcare is also covered in this survey paper. The goal of this paper is to provide a comprehensive survey of recent advancements in data augmentation techniques and their application in various domains.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → **Neural networks**; **Image processing**;

Additional Key Words and Phrases: Deep learning, data augmentation, transfer learning, cost sensitive learning, generalization, overfitting

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1 INTRODUCTION

Deep learning has made incredible progress in various practical applications. Recent advancements in deep learning [Xizhao Wang et al. 2020] include the advancement of deep neural architecture, powerful computation, and access to big data, which has increased its value in real-life applications. It is used to develop a model that works like a human or even better in different application domains. It covers a different set of practical applications such as face detection [Guo and Zhang 2019], pedestrian detection [Brunetti et al. 2018], automatic machine translation [Costa-jussà et al.

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2017], speech recognition [Fayek et al. 2017], natural language and image processing [Iqbal and Qureshi 2020], predictive forecasting [Sezer et al. 2020], and used even in highly advanced applications such as in self-driving cars [Fujiyoshi et al. 2019] and also in the healthcare domain [Dai and Wang 2018]. However, building a deep learning model has its own set of challenges. Generalization [Neyshabur et al. n.d.] refers to the capability of a model to recognize new unseen data. Generalization is one of the major challenges while building a deep learning model. A model with poor generalization usually overfits the training data. Overfitting [Karystinos and Pados 2000] is a modelling error that occurs when a model tries to fit all the data points available in the training dataset. A robust model requires a lot of reliable data for training the model. If a model has been trained on a limited set of useful data, it will be unable to generalize accurately. Even though it can make accurate predictions for previously seen training data, whenever tested for some new data it will make inaccurate predictions, making the model useless. To reduce the problem of model overfitting and to improve generalization performance, the model requires more training dataset as more data make the model unable to overfit all the samples. In some industries collecting data is either not feasible or requires more resources. In the medical field, data is not shared because of privacy concerns. A lot of data is required in the field of healthcare, research, video surveillance, and also to develop autonomous things such as robots and self-driving cars. The process of data collection demands a lot of time and money. One of the data space solutions to the problem of limited data is data augmentation [Junhua Ding et al. 2019]. This method is used to artificially generate data from the available dataset. It creates new data by transforming the already existing dataset, so there is no need to collect new data. It increases the amount and variety of data in the datasets to train and test the model. Data can be augmented either by learning a generator that can create data from the scratch that can be done by GAN networks [Alqahtani et al. 2019] or by learning a set of transformations that can be applied to already existing training set samples [Cubuk Zoph, Vasudevan et al. n.d.] to improve the performance of deep learning models.

1.1 Motivation for Work

The inspiration of this survey is to thoroughly demonstrate various data augmentation techniques to deal with the sample inadequacy problem while training a deep learning model for different real-life applications and to provide an overview of data augmentation applications in various domains. This article proposes a cross-wise view of the present trends in the data augmentation techniques and their comparative analysis. The distinct motivation for this comprehensive survey is as follows:

- (a) To study existing and efficient techniques of data augmentation for dealing with data inadequacy in various application domains.
- (b) To analyse the applicable areas of data augmentation and to demonstrate how the performance of real-world applications improved by using augmented data.

In order to understand the need for data augmentation and its applications in various domains, it's important to explore various existing augmentation techniques applicable to different application areas. It will help to propose new methods to deal with the data inadequacy problem and to improve the performance of generalization.

1.2 Our Contributions

A comprehensive review has been conducted to investigate various data augmentation techniques for improving the performance of deep learning models. In Section 6, augmentation taxonomy is categorized into two parts: image processing to augment data and training neural networks so that they can learn optimal policies to augment data. Research Methodology in Section 2 is

designed to study and compare various data augmentation techniques. This is done using SLR based on general guidelines proposed by [Kitchenham and Brereton 2013]. Data augmentation helps in solving the problem of data scarcity, but there are also other methods that deal with the issue of limited dataset as discussed in Section 5. These include transferring knowledge or using a cost-sensitive approach to deal with the issue of imbalance dataset. In the Section 6 of this paper, all existing data augmentation methods and their limitations are summarized in tabular format. Existing techniques are also evaluated on different benchmark datasets with the help of bar graphs. We also present a listing of data augmentation applications in different domains in Section 8. These applications can be further divided into four categories: computer vision, natural language processing, security, and healthcare.

1.3 Article Organization

Section 1 presents an introduction to the research work related to data augmentation and the motivation behind it. Section 2 represents the schematic representation of **Systematic Literature Survey (SLR)**. Section 3 represents related literature survey and Section 4 represents issues and challenges identified from literature review. Section 5 describes other methods to solve the problem of data scarcity. Section 6 describes various approaches for data augmentation for different domains. A comparative analysis of augmentation techniques is presented in Section 7. Section 7 also presents the benchmark datasets on which augmentation successfully works with their evaluation metrics. Section 8 describes application domains of data augmentation and related literature to applications. Section 9 presents our conclusion and in Section 10 future scope of data augmentation is given.

2 RESEARCH METHODOLOGY

Research Methodology intends to study and compare different data augmentation techniques. It is designed by using SLR based on the general guidelines proposed by [Kitchenham and Brereton 2013]. Approximately 100 research papers from reputed journals and professional conferences are reviewed. Figure 1 explains the protocol that is used to carry out this SLR.

The key steps used to design this SLR are listed below:

- Identifying research questions to design SLR. (Section 2.1)
- Listing distinct keywords to search research papers related to research questions. (Section 2.2)
- Applied inclusion and exclusion to filter out research papers that fit in the domain. (Section 2.3)
- Performed backward and forward chaining to search relevant literature. (Section 2.4)
- Results from different research papers is used for future research. (Section 2.5)

2.1 Aim and Research Questions/Research Questions (RQ) Identification

This SLR aims to answer the following **research questions (RQs)**:

- RQ1: How can existing research on data augmentation be classified.
- RQ2: What patterns, gaps, and challenges could be inferred from the current research efforts that will help in future research.
- RQ3: What is the significance of data augmentation techniques in improving performance of deep learning models.
- RQ4: What is the contribution of data augmentation in various real-world problems.

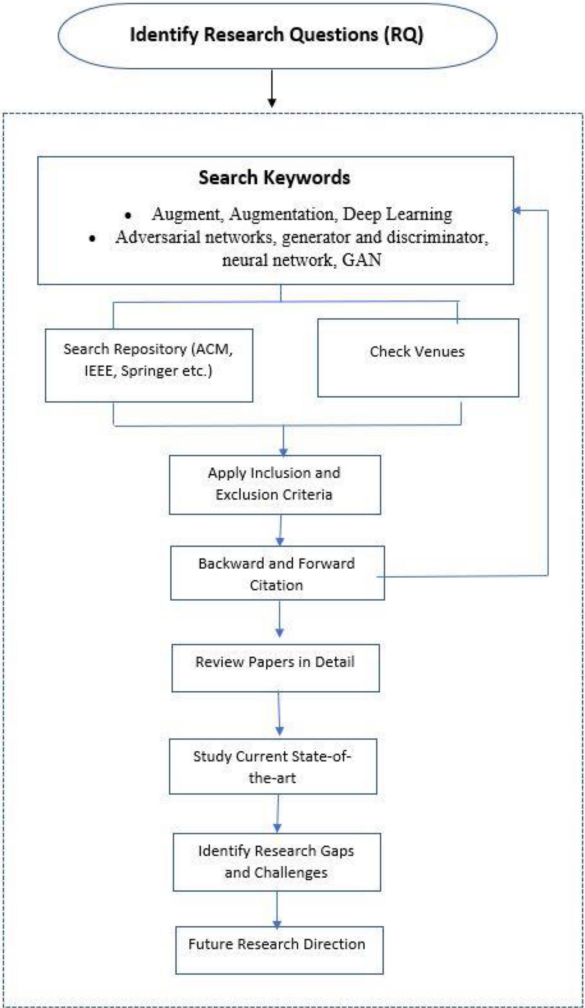


Fig. 1. Schematic representation of SLR.

Table 1. Search Keywords Specific to the Research Domain

Research Domain (R)	Keywords (K)
Data Augmentation	Augment, Augmentation, Deep Learning, Data
Technical Approach	Autoencoders, GAN, Adversarial Networks
Classification	Image Classification, Classification Techniques
Generative Adversarial Networks (GANs)	Adversarial Networks, Generator and Discriminator, Neural Network
Learning Models	Deep Learning, Machine Learning

2.2 Search Strategy

Search strategy to find relevant literature related to research questions are discussed below:

- Search Keywords. The aim is to list search keywords to find relevant literature. Listed keywords are searched on the title, abstracts, and meta-data like tags on the research papers. Table 1 lists keywords related to the research domain.
- Search Repositories and Datasets. We have considered reputed repositories such as ACM Digital Library, Science Direct, IEEE Xplore and Springer Link to find relevant publications related to our domain. Table 2 list the top publications considered in this survey:

Table 2. Top Publications Studied with their H-index Values

Sr. No.	Publications	H-index
I	International Conference on Learning Representations	203
II	Neural Information Processing Systems	198
III	AAAI Conference on Artificial Intelligence	126
IV	Expert Systems with Applications	111
V	IEEE Transactions on Neural Networks and Learning Systems	107
VI	Neurocomputing	100
VII	Applied Soft Computing	96
VIII	Knowledge-Based Systems	85
IX	Neural Computing and Applications	67
X	Neural Networks	64

Figure 2 represents top 10 publications with their H-index values:

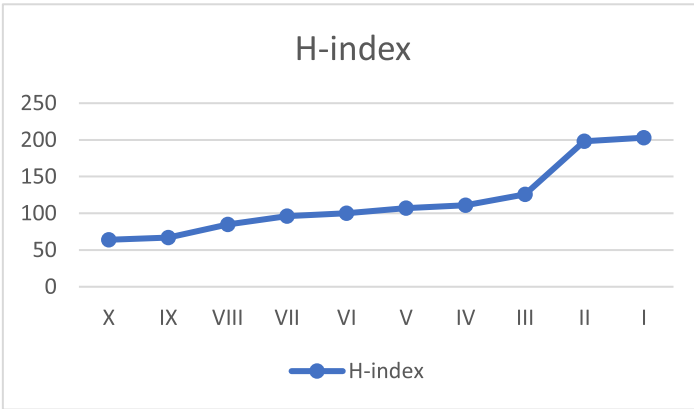


Fig. 2. H-index value of top publications.

2.3 Selection Criteria

Selection Criteria is set to filter out relevant literature as all retrieved papers are not under the scope of this SLR. For selecting relevant literature, inclusion and exclusion criteria are used.

- Inclusion Criteria. We have collected the papers from 2010–2021. We have included all the retrieved papers that are related to data augmentation techniques.
- Exclusion Criteria. Exclusion criteria followed in this SLR are listed below:
 - Short length papers are rejected as most of them are preliminary work.
 - Papers written in English language are considered over other languages as it is a common language used by reviewers and researchers.



Fig. 3. Word Cloud of titles of research papers studied.

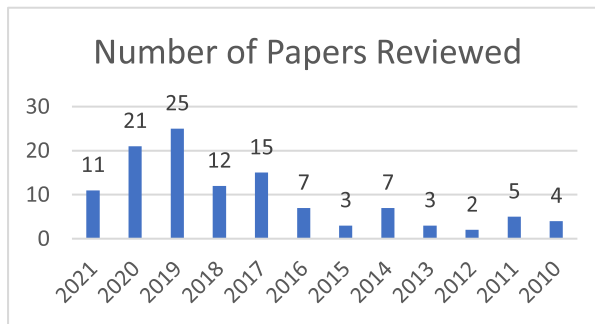


Fig. 4. Year wise representation of number of papers studied.

- Papers published before 2010 are not considered in this SLR.

2.4 Backward and Forward Citation

To expand the research, citation chaining or reference mining methods are used. It helps in retrieving additional relevant papers related to our domain by reviewing cited papers. Figure 3 represents a word cloud of titles of research papers which can be useful for retrieving relevant research papers. Research papers can be traced in backward as well as in the forward direction.

- Backward Chaining: It helps in identifying existing resources regarding the same topic.
- Forward Chaining: It helps in identifying those papers that cite the existing resources.

2.5 Research Publication Selection

Research publications selected are represented via chart. Figure 4 represents year-wise selected articles with paper count.

Initially, papers are searched based on keywords defined in Section 2.2, then to further filter out relevant literature selection criteria is considered as discussed in Section 2.3. Based on citation chaining as discussed in Section 2.5, additional work is retrieved.

3 LITERATURE REVIEW

Data scarcity is a major issue while building a deep learning model, as in many fields a sufficient amount of data is not available to train the model. Data augmentation can help us to deal with this problem, and many researchers have contributed in this area. In this section, we present the contributions of other researchers in this domain with the help of Table 3 as mentioned below:

Table 3. Research Table of Papers Studied

Paper Title	Objective	Methodology	Findings
Deep Image: Scaling up Image Recognition [Wu 2014]	To build a supercomputer using large deep neural networks and multi class high resolution images	Trained a large convolutional neural network with multi scale images and also on down sampled images to compare their performance. Used data augmentation by color casting to alter the intensities of the RGB channels in training images.	Image Recognition accuracy improves by using high resolution images as in the case of downsized images, it loses too much information
Improved Regularization of Convolutional Neural Networks with Cutout [DeVries and Taylor 2017b]	To present a simple regularization technique called as cutout can be used to improve the robustness and overall performance of CNN	Region of an image is cutout or it is randomly masking out regions of input training image.	Cutout can be used as a data augmentation technique to solve the problem of data inadequacy and can improve model robustness
Forward Noise Adjustment Scheme for Data Augmentation [Moreno-Barea et al. 2019]	To propose a new method for data augmentation i.e., by adding noise to input images	Proposed work injects a matrix of random values usually obtained from Gaussian distribution to improve prediction accuracy in classification problems.	It improves prediction accuracy in classification problems and can be used for supervised training on deep learning architectures
Data Augmentation by Pairing Samples for Images Classification [Inoue 2018]	To propose a simple and effective approach for data augmentation called as sample pairing	In this approach two images undergo different image processing methods, firstly two images are randomly cropped and then randomly flipped horizontally. Then the images are mixed by averaging the pixel values for each RGB channel. The label of new mixed image is same as the first randomly selected image.	This technique yields significant improvement in accuracy for tasks such as classification and reduces the problem of overfitting
Data augmentation for improving deep learning in image classification problem [Mikołajczyk and Grochowski 2018]	To propose an approach for improving deep learning in the task of image classification, and also to compare various data augmentation techniques	In order to improve training process, it pre-trains the neural network with newly created images. It generates new images by combining the content of a base image with the appearance of another image. Proposed method is validated on the three medical cases, which utilize image classification for the diagnosis.	Proposed method improves the performance of deep learning model
Overfitting Mechanism and Avoidance in Deep Neural Networks [Salman and Liu 2019]	To propose an algorithm to avoid the problem of overfitting to improve the accuracy of classification tasks especially when the number of training dataset is limited and to demonstrate the concept of generalization	Proposed a consensus-based overfitting avoidance algorithm that allows model to identify samples that are classified due to random factors using multiple models. They have also showed how to avoid overfitting after identifying the overgeneralized samples based on the training dynamics.	Proposed work improves the performance of classification task and also reduce the problem of overfitting
Comparison of Traditional Transformations for Data Augmentation in Deep Learning of Medical Thermography [Ornek and Ceylan 2019]	To compare traditional transformations used for data augmentation in deep learning	By using neonatal thermal images, they compared various traditional data augmentation methods such as rotating, mirroring, zooming, shearing, histogram equalization, color changing, blurring, sharpening, and brightness enhancement. These traditional methods are used in the classification of medical thermo grams.	The performance of classification increased the accuracy rate by 26.29%.

(Continued)

Table 3. Continued

Paper Title	Objective	Methodology	Findings
Improved Mixed-Example Data Augmentation [Summers and Dinneen 2019]	To explore the domain of mixed image space for data augmentation	Proposed work explores various linear (Mixup and BC+) and non-linear methods (Vertical Concat, Horizontal Concat, Mixed Concat, Random 2*2, VH Mixup, VH BC+, Random Square, Random pixels and Noise Mixup) to mix images. VH BC+ non-linear method performs remarkably well.	Explored the domain of mixed image space, to see that linearity is important for mixing images or not and they came across various nonlinear methods to mix images which surprisingly give better accuracy and improves generalization
Adversarial Framing for Image and Video Classification [Zajac et al. 2019]	To use adversarial framing approach for classification task in both image and video dataset	Proposed a method that adds an adversarial framing on the border of the image and rest keeps the image remain unchanged. It limits the adversarial attack just limited to the border of images. It helps the network to learn augmentations that results in misclassification and help in forming an effective algorithm.	Proposed method only adds small border around image and does not modify original content of the image. It can be used as a data augmentation technique for the classification task
Data augmentation using generative adversarial networks for robust speech recognition [Qian et al. 2019]	To propose a new framework for robust speech recognition using generative adversarial networks	Synthetic data generated frame by frame based on spectrum feature level by using basic GAN. There are no true labels existed for them and this unsupervised learning framework is used for acoustic modeling. For better data generation conditional GAN is used which explored two different conditions to provide true labels directly. Then during acoustic model training these true labels are combined with soft labels for improving the performance of the model.	Proposed method improves the performance of speech recognition task
A survey on face data augmentation for the training of deep neural networks [Xiang Wang et al. 2020]	To study various data augmentation approaches applied for face related tasks	It gives an outline of how to do face augmentation and what it can do. They perform different face data transformations which includes geometric and photometric transformation, hairstyle transfer, facial makeup transfer, accessory removal or wearing, pose transformation, expression synthesis and transfer, age progression and regression, and some other types of transformations to enrich the face dataset.	All the augmentation methods used in this paper can be used to improve the robustness of the model by increasing the variation of training data
Object-adaptive LSTM network for real-time visual tracking with adversarial data augmentation [Du et al. 2020]	To propose an object - adaptive LSTM network for real time visual tracking with adversarial data augmentation	LSTM network fully exploits the sequential dependencies and helps to effectively adapt to the object appearance variation in a complex scenario. They also used matching based tracking method for selection of high-quality dataset to feed it to the LSTM network. To solve the problem of sample inadequacy and class imbalance they used GAN to create augmented data which facilitates the training of the LSTM network.	Proposed method robustly tracks an arbitrary object without the risk of overfitting

(Continued)

Table 3. Continued

Paper Title	Objective	Methodology	Findings
A novel data augmentation scheme for pedestrian detection with attribute preserving GAN [Songyan Liu et al. 2020]	To propose a data augmentation approach for pedestrian detection	This approach helps to tackle the problem of insufficient training data coverage by transferring the source pedestrians to a target scene. Then transferring its style by APGAN (Attribute Preserving Generative Adversarial Networks).	Proposed work helps by providing variation in dataset and proposed method yields significant results to improve the generalization ability of the detector and enhance its robustness
A multi-cascaded model with data augmentation for enhanced paraphrase detection in short text [Shakeel et al. 2020]	To propose a data augmentation strategy and a multi cascaded model for paraphrase detection	The augmentation method generates paraphrase and non-paraphrase annotations based on graph analysis of existing annotations. The multi cascaded model employs multiple feature learners to encode and classify short text pairs.	This approach yields significant improvement results of deep learning models for paraphrase detection

Many researchers have proposed methods to augment data in many domains and apply these methods on different datasets. They have also demonstrated the relation between data and the deep learning model. [Weiss et al. 2016] proposed a review on transfer learning that can transfer knowledge from the source domain to target domain to predict future outcomes when we have a limited set of data to build a deep learning model. [Shorten and Khoshgoftaar 2019] presented a survey paper on various image data augmentation techniques which can improve the performance of deep learning models. Data augmentation yields significant results fields in various domains, from image classification [D. Han et al. 2018] to speech recognition [Qian et al. 2019] and improves model overall performance. Data can be generated by simple affine transformations. Combination of these transformations can improve performance in some specific domains [Ratner et al. n.d.]. For example, [Krizhevsky et al. n.d.] augmented ImageNet dataset for image classification by combining various affine techniques i.e., by translation, horizontal reflection, and altering values of RGB pixels. Adversarial training [Goodfellow et al. 2014] is another class for data augmentation. Adversarial examples are used to enlarge the dataset [Szegedy et al. 2013], and by training the model with these adversarial examples increase the robustness of models [Bastani et al. n.d.; Carlini and Wagner 2017; Goodfellow et al. 2014; Szegedy et al. 2013]. GAN augmentation [Goodfellow et al., n.d.] is widely used to generate synthetic images and it is applicable in various domains [Bang et al. 2020; Chae et al. 2019; Lu et al. 2019; Pandey et al. 2020], to improve GAN architecture many methods are proposed [Salimans et al. n.d.] which help to generate accurate synthetic dataset. Further various meta-learning approaches [Cubuk Zoph Vasudevan et al., n.d.; Lemley et al. 2017; Perez and Wang 2017] are proposed to augment data and improve the performance of deep learning models.

Table 4 represents different deep neural networks used for augmentation with their number of papers reviewed in different publications.

4 ISSUES AND CHALLENGES

• Limited Training Data

In many application domains, there is a limited set of data available for training of neural networks. In some industries collecting data is either not feasible or requires more resources. In the medical field, data is not shared because of privacy concerns. A lot of training data is required in the field of marketing, research, video surveillance, and also to develop autonomous things such as robots, self-driving cars, etc. It requires a lot of time and money

Table 4. Research Papers Reviewed for Various Deep Neural Networks

Network	Publications	#Papers
CNN	Gatys et al. [2015], Wu et al. [2015], Masi et al. [2016], Wang et al. [2017], Lemley et al. [2017], Zhong et al. [2017], Devries et al. [2017], Shijie et al. [2017], Krizhevsky et al. [2017], Bowles et al. [2018], Inoue et al. [2018], Taylor et al. [2018], Mikolajczyk et al. [2018], Salman et al. [2019], Omek et al. [2019], Takahashi et al. [2019], Jackson et al. [2019], Qian et al. [2019], Wang et al. [2020], Liu et al. [2020], Shakeel et al. [2020], [Mushtaq et al. 2021], [Hidayat et al. 2021], [Agarwal et al. 2021]	24
GAN, CGAN	Wang et al. [2017], Shijie et al. [2017], Bowles et al. [2018], Mikolajczyk et al. [2018], Qian et al. [2019], [Cheng 2019], Qian et al. [2019], Wang et al. [2020], Du et al. [2020], [P. Wang et al. 2020], Liu et al. [2020], [X. Pan 2021], [Z. Zhu et al. 2021], [Andresini et al. 2021],	14
ResNet	Sun et al. [2017], Zhong et al. [2017], Zoph et al. [2019], Summers et al. [2019], Zajac et al. [2019], [Yulin Wang et al. 2021]	6
RNN, LSTM	Cubuk et al. [2019], Devries et al. [2017], Du et al. [2020], Shakeel et al. [2020], [Katiyar & Borgohain, 2021], [Sisi Liu et al. 2020]	6

to collect more data. One of the data space solutions to the problem of limited data is data augmentation. Data augmentation is a technique used to artificially generate data from the available dataset. It saves cost and time consumption to collect new data and reduces the problem of sample inadequacy in deep learning models.

- **Lack of Relevant Data**

Training a deep learning model requires a large amount of relevant data to improve its performance. Data augmentation can be done by various techniques that can enhance the size and quality of training datasets to build better deep learning models as discussed in the literature review.

- **Model Overfitting**

For practical applications, the deep learning model must be reliable so that it can generalize properly. These models require a large amount of data to avoid the problem of overfitting. Overfitting is a modelling error that occurs when a model too closely fits to the available dataset. If a model has been trained on a limited set of useful data, it will be unable to generalize accurately for a new set of data. Even though it can make accurate predictions for the training data but whenever these models are tested for some new data it will make inaccurate predictions, making the model useless. To reduce the problem of overfitting and to improve generalization performance, the model requires more datasets. Data augmentation reduces the problem of overfitting by training the model with a large amount of relevant data. It regularizes the model and improves its capability of generalization.

- **Unbalanced Dataset**

Deep learning models require lots of data of each class to classify accurately but sometimes data available is imbalanced which makes it difficult to train the deep learning model and affects its overall performance. Data imbalance is a major issue faced while dealing with real life applications. Data can be resampled to deal with the imbalance problem but augmenting data can help in dealing with the imbalance dataset by creating more data for training the deep learning model.

5 METHODS TO DEAL WITH A LIMITED DATASET

In this section, we address the problem of data scarcity and some solutions that can be useful to deal with a limited dataset. If the target domain has limited data, then we can either transfer

knowledge from the related source domain [S. J. Pan and Yang 2010] or can generate synthetic data [Lei et al. 2019] via augmentation techniques. In particular, we would like to cover the following topics:

I. Transfer knowledge

In this part, we address the issue of data scarcity and study how we can transfer knowledge from models or we can borrow knowledge from the experts. It improves the model by transferring knowledge from the related source domain to target domain to predict future outcomes. It can be achieved by sources mentioned as below:

(a) From Models

- In cases of limited data, where collecting and labelling data is expensive or the data is inaccessible, there we can transfer knowledge from one model to another. In training deep learning model, if two models are related to some domain, then we can transfer knowledge to improve the results of the target learner.
- [Weiss et al. 2016] provides a comprehensive review on homogeneous transfer learning, heterogeneous transfer learning, and negative transfer learning, and about the applications of transfer learning. Transfer learning work successfully in many application domains such as image recognition [W. Li et al. 2014; Y. Zhu et al. n.d.], human activity classification [Harel and Mannor 2011], multi-language text classification [Prettenhofer and Stein 2010; Zhou et al. n.d.], and software defect classification [Nam et al. 2018].
- Homogeneous transfer learning is applicable where input feature space is the same for both source and target domain. It includes instance-based transfer learning [Apte et al. 2011], [Yao and Doretto 2010], asymmetric feature-based transfer learning [Daumé III 2007; Duan et al. 2012; M. Long et al. 2014], symmetric feature-based transfer learning [Oquab et al. 2014; S. J. Pan et al. 2011], parameter-based transfer learning [Tommasi et al. 2010; Yao and Doretto 2010], relational-based transfer learning [F. Li et al. 2012], and hybrid-based transfer learning [Xia et al. 2013].
- Heterogeneous transfer learning [Day and Khoshgoftaar 2017] is applicable when the feature space of the source and target domain is different. It includes symmetric [F. Li et al. 2012] and asymmetric [Kulis et al. 2011] approach of transfer learning.
- When the source domain is not related to the target domain, then the target learner can be negatively impacted because of the weak connection between source and target domain, and such learning is termed as negative learning [Seah et al. 2013].

(b) From domain expert

- In this part, we address the issue of limited data and to deal with it we borrow knowledge from external domain experts [Shi et al. n.d.]. The main challenge is to transfer different format knowledge to a learning model.
- Enriching Transformations using knowledge graph: Knowledge graph can be used to transfer knowledge in various application domains. For example: the health domain knowledge graph can be used for diagnosis of health related issues [Choi et al. 2017; Ma Chitta et al. 2018].
- Regularizing the loss function by incorporating domain knowledge: loss function can be regularized to transfer domain knowledge or constraint can be added in the loss function [Ma You et al. 2018].
- Many researchers focus on improving the performance of the model by using a knowledge graph [X. Han et al. n.d.; Malik et al. 2020; J. Zhang et al. n.d.; Zhao et al. 2020].

II. Cost-Sensitive Learning

Class Imbalance is one of the challenging problems while training deep learning models. In real life problems, data collected is not balanced and class with majority overwhelm the classifier which result in having a high false negative rate. To deal with imbalance data we can either resample the data or can apply cost sensitive learning. Cost sensitive learning can solve the issue of imbalance dataset by assigning misclassification cost to each class, so that instead of optimizing accuracy, the problem is then to minimize the total misclassification cost.

- [Khan et al. 2018] proposed a CoSen deep CNN architecture to deal with the problem of class imbalance. Proposed CoSen can automatically learn robust feature representations for both the classes (Majority and Minority classes) and can be applicable to both binary and multiclass problems.
- Cost sensitive learning can be applicable in various applications with imbalance dataset: [Aceto et al. 2019] tackled mobile (encrypted) traffic with a deep learning approach with cost sensitive learning and termed it as MIMETIC. [Olowookere and Adewale 2020] proposed a framework that combines meta-learning ensemble techniques and cost sensitive learning for fraud detection.

III. Data Augmentation

Augmenting the available training datasets gives remarkable results in improving the performance of the model by improving model generalization and by reducing the problem of generalization. Overfitting can be regularized in many ways such as dropout [Srivastava et al. 2014], batch normalization [Ioffe and Szegedy n.d.], zero-shot learning [Palatucci et al. n.d.], and transfer learning [S. J. Pan and Yang 2010; Shao et al. 2015], but data augmentation deals with the main challenge of building a model i.e., data.

- Data augmentation can be done online or offline. In online augmentation, data is augmented at training time so that there is no need to store the augmented data [Lemley et al. 2017]. In offline augmentation data is augmented in the pre-processing phase and stored on the disk [Perez and Wang 2017].
- Data augmentation can be achieved by various approaches such as a heuristic approach [Ratner et al. n.d.], adversarial approach [Goodfellow et al. 2014], style transfer approach [Gatys et al. 2016], or by selecting the best optimal policy [Cubuk, Zoph, Vasudevan et al., n.d.].
- It helps in improving model performance in various application domains. A lot of work has been proposed for image classification. Recent advancements in augmentation include meta-learning [Cubuk, Zoph, Vasudevan, et al., n.d.; Lemley et al. 2017; Perez and Wang 2017] and GAN network [Goodfellow et al., n.d.; Lu et al. 2019] which help to extend its usage in the field of computer vision [Chen et al. 2017; Jun Ding et al. 2016; Du et al. 2020; Songyan Liu et al. 2020; Meng et al. 2019; Yong Wang et al. 2020], natural language processing [Sisi Liu et al. 2020; Y. Long et al. 2020; Shakeel et al. 2020], healthcare domain [Frid-Adar et al. 2018; Sajjad et al. 2019] and in dealing with the problem of class imbalance [Johnson and Khoshgoftaar 2019].
- Various methods are proposed to augment data, some of them are domain-specific [Summers and Dinneen 2019; Takahashi et al. 2019] and some are evaluated on different input domains [DeVries and Taylor 2017a]. Some traditional and recently proposed data augmentation techniques are discussed in Section 6. Comparative analysis of existing techniques is also presented in Section 7.

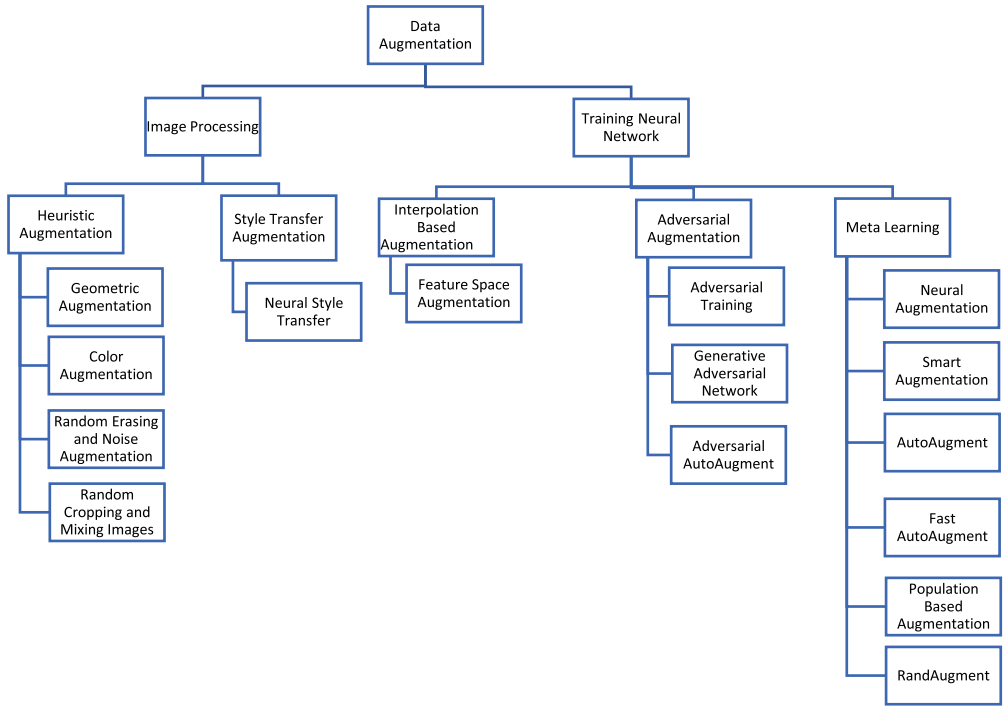


Fig. 5. Taxonomy of Data Augmentation.

6 DATA AUGMENTATION

Data augmentation is used to generate artificial data by transforming available training datasets to build a deep learning model. It can be achieved by various methods from the traditional approach to making the model learn basic transformations. Figure 5 represents the taxonomy of data augmentation. Most of the earlier approaches are domain-specific that are applicable only on a defined dataset domain, but recent advancements deal with proposing methods that can be applicable to different dataset to increase its efficiency, so that it can be used in various application domains and improve the capability of the model to generalize accurately. Table 5 represents various augmentation techniques according to their augmentation approaches. Some of the augmentation methods are discussed in this section followed by the table as given below:

(a) Geometric Augmentation

Geometric transformations are traditional methods to generate data artificially. These are based on basic image transformation techniques. Some of the important geometric transformations are listed below:

- Flipping is a mirror effect, done by reversing the pixels of an image horizontally or vertically. This data augmentation proved useful on image datasets such as CIFAR10 and ImageNet.
- Cropping is used to create image data with mixed width and height dimensions. Random cropping gives a similar effect to translations.
- Rotation augmentation is done by simply rotating the image at a certain angle. It is useful for image data augmentation but in the case of digit data such as MNIST, only slight rotations are useful.

Table 5. Comparison of Various Augmentation Techniques

Sr. No.	Title	Objective	Dataset Used	Observation	Limitation
1.	Image Style Transfer Using Convolutional Neural Networks [Gatys et al. 2016]	To propose data augmentation method using artistic approach i.e., by neural algorithm	Content image: Neckarfront in Tübingen, Germany Style reference image: (1) The Shipwreck of the Minotaur, (2) The Starry Night, (3) Der Schrei, (4) Femme nue assise, and (5) Composition VII.	Neural algorithm with style transfer is a creative approach which improves the performance of models by creating images for different visual environment.	Domain Specific technique. It is only applicable to image dataset.
2.	Generative Adversarial Nets [Goodfellow et al. n.d.]	To propose a generative model via an adversarial approach which includes training of two models simultaneously: Generator and Discriminator	MNIST, Toronto Face Database (TFD), CIFAR-10	Better than other generative models as no inference is needed while training and representation of adversarial network is very sharp, even degenerate distributions.	Due to unstable training and unsupervised learning method, it becomes harder to train and generate output.
3.	Dataset Augmentation in Feature Space [DeVries and Taylor 2017a]	To present a domain independent augmentation technique for training supervised learning models to improve its overall performance	UJI Pen Characters dataset, Arabic Digits dataset, Australian Sign Language Signs dataset, UCF Kinect action recognition dataset, MNIST and CIFAR-10	Proposed technique is tested on five datasets from different domains. The amount of variability within the dataset increases by using this technique.	Extrapolation generates useful data when used in feature space. Interpolation tends to tighten class boundaries and lead to overfitting.
4.	The Effectiveness of Data Augmentation in Image Classifications using Deep Learning [Perez and Wang 2017]	To use meta learning approach for data augmentation i.e., to help the neural net to learn augmentation	Tiny-imagenet-200 data, MNIST	Proposed meta learn approach i.e., neural augmentation, reduces the problem of overfitting via data augmentation and also improves the classifier.	Domain Specific technique. It is only applicable to image dataset.
5.	Smart Augmentation-Learning an Optimal Data Augmentation Strategy [Lemley et al. 2017]	To introduce a new method to reduce overfitting i.e., by smart augmentation. They do not address any manual augmentation nor does their network attempt to learn simple transformations, the only aim is to learn the best approach for a given set of inputs	AR faces dataset, FERET, Adience, MIT Places	Proposed method creates a network that learns how to generate augmented data during the training process to reduce overall network loss. In this paper, proposed method is used for gender recognition, in contrast to traditional augmentation techniques it improves the performance of model and increases its accuracy from 88.15% to 89.08% and reduces overfitting.	Smart Augmentation achieves better results when used in small networks rather than using larger networks.

(Continued)

Table 5. Continued

Sr. No.	Title	Objective	Dataset Used	Observation	Limitation
6.	AutoAugment: Learning Augmentation Strategies From Data [Cubuk, Zoph, Vasudevan, et al., n.d.]	To develop a new procedure to automatically search for improved data augmentation policies termed as AutoAugment	CIFAR-10, CIFAR-100, SVHN, Stanford Cars, ImageNet	Proposed work is an effective technique for data augmentation by searching best policies of augmentation.	High Computational cost and AutoAugment spend most of its time in search of optimal policy. As search speed is very slow, it is time consuming.
7.	Fast AutoAugment [Lim et al. n.d.]	To propose an algorithm to search best augmentation policies in more efficient ways by using searching based on density matching between a pair of train datasets	CIFAR-10, CIFAR-100, SVHN, ImageNet	Fast Autoaugment in comparison to Autoaugment speeds up the search time to find best policy.	High Computational cost and results of FastAutoaugment and Autoaugment technique are similar, no such improvement as expected.
8.	Population Based Augmentation: Efficient Learning of Augmentation Policy Schedules [Ho et al. n.d.]	To propose an algorithm called as population-based augmentation (PBA) which is helpful in choosing an effective augmentation strategy from a large search space	CIFAR-10, CIFAR-100, SVHN	PBA trains and optimizes a series of population of neural network parallel with random hyper parameter and finds best optimal state quickly.	In PBA technique, there is a slight real time overhead because of parallelization as new augmentation policies can only be trained after completion of previous batch.
9.	Randaugment: Practical Automated data augmentation with a reduced search space [Cubuk, Zoph, Shlens, et al. n.d.]	To improve earlier introduced automated augmentation strategy	CIFAR-10, CIFAR-100, SVHN, ImageNet, COCO dataset	Earlier a search for both magnitude and probability of each operation is done independently for each proxy task, now to reduce computational expense a simplified search space is proposed to search a single distortion magnitude that jointly controls all operation without any separate search space for proxy task.	Results are limited to some benchmark datasets and not tested for other domains related to text and speech.
10.	Adversarial AutoAugment [X. Zhang et al. n.d.]	To propose an adversarial method to automate augmentation and termed it as Adversarial AutoAugment	CIFAR-10, CIFAR-100, ImageNet	Proposed method tries to increase the training loss of target network by generating adversarial policies for augmentation so that the target network can learn more robust features in order to improve generalization.	High computational cost while training target network but overall cost is less than AutoAugment technique.

- Translation is to shift the original image in a direction i.e., right, left, up or down, very useful to preserve the label. After translation of an image in a particular direction, the remaining space is padded to preserve spatial dimensions.

Geometric transformations are easy to implement but require additional memory and additional training time. In some transformations, it requires manual observations to ensure label preserving. Therefore, the scope of geometrical transformations is relatively limited.

(b) Color Augmentation

Color space transformation is also known as photometric transformation. These types of transformations can be easily done by image editing apps. Like geometric transformation, it also requires additional memory and training time. [Taylor and Nitschke 2019] compared the effectiveness of geometric and photometric transformations in the input dataset.

(c) Random Erasing and Noise Augmentation

Random erasing introduced by [Zhong et al. n.d.] is another data augmentation technique. In this, it randomly selects a rectangle region of an input image and erases its original pixels with random values. It is inspired by the mechanisms of dropout regularization [DeVries and Taylor 2017b]. This technique was designed to overcome the image processing challenges due to occlusion. Noise injection is to inject a matrix of random values usually obtained from Gaussian distribution to improve prediction accuracy in classification problems. Noise injection is tested by [Moreno-Barea et al. 2019] on nine datasets from the UCI repository and used for supervised training on deep learning architectures.

(d) Random Cropping and Mixing Images

[Inoue 2018] proposed a simple approach to augment data. In this approach, two images undergo two different image processing methods. Firstly, two images are randomly cropped and then randomly flipped horizontally. Then the images are mixed by averaging the pixel values for each RGB channel. The label of the new mixed image is the same as the first randomly selected image. Further, this approach is investigated by [Summers and Dinneen 2019], they used non-linear methods to combine images. [Takahashi et al. 2019] proposed another approach for mixing images by randomly cropping images and concatenating the cropped images to form new images.

(e) Feature Space Augmentation

[DeVries and Taylor 2017a] used a domain-independent augmentation technique for training supervised learning models to improve overall performance. They trained a sequence auto-encoder to construct a learned feature space in which they extrapolate between samples. The amount of variability within the dataset increases by using their technique. In their paper, they demonstrated their technique on five datasets from different domains i.e., speech, motion capture, sensor processing, and images.

(f) Adversarial Training

It attempts to fool models by providing malicious input. A network that is trained on adversarial examples is one of the few defences against attacks in deep models such as adversarial attacks. [Zaj et al. n.d.] proposed a method that adds an adversarial framing on the border of the image and the rest keeps the image remaining unchanged. They used the adversarial framing approach for classification tasks in both image and video datasets which limits the adversarial attack just to the border of images. It helps the network to learn augmentations that result in misclassification and helps in forming an effective algorithm.

(g) GAN Augmentation

GANs were first introduced by [Goodfellow et al. n.d.] and used for effective data augmentation. GAN is a type of generative model i.e., it can produce new content based on its available training data. A GAN is made up of two **Artificial Neural Networks (ANNs)** that work against each other, termed as Generator and Discriminator. The first one creates new data instances from available data while the second evaluates the generated data for authenticity. GANs are very useful in different fields, also in the healthcare domain [Frid-Adar et al. 2018].

(h) Neural Style Transfer

Neural Style Transfer [Gatys et al. 2016] is one of the artistic approaches for data augmentation. Neural style transfer is an optimization technique that blends two images to create a fine new image, it defines three images: a content image (the image we want to transfer a style to), a style reference image (the image we want to transfer the style from such as an artwork by a famous painter), and the input (generated) image. It blends them together such that the input image is transformed to look like the content image, but “painted” in the style of the style image.

(i) Neural Augmentation

[Perez and Wang 2017] presented an algorithm to meta learn a neural style transfer technique termed as neural augmentation. This method helps the neural net to learn augmentations. There are two parts of the network in the training phase. The augmentation network takes two random images from the training set and produces a single image. Then the original image with new image is fed to the classifying network. The training loss is then back propagated to train the augmenting layers of the network as well as the classification layers of the network.

(j) Smart Augmentation

[Lemley et al. 2017] introduced a new method to reduce overfitting, termed as Smart Augmentation. It creates a network that learns how to generate augmented data during the training process to reduce overall network loss. The main aim of Smart Augmentation is to learn the best approach for a given set of inputs. It uses two networks: Network-A and Network-B. Network-A is considered as an augmentation network, it uses a series of convolutional layers that takes two or more input images and maps them to create a new image or images to train Network-B. Any change in error rate in Network-B is back-propagated to Network-A to update it. It was tested for the gender recognition tasks. This was compared with traditional augmentation techniques and as a result it was noted that the accuracy increased from 88.15% to 89.08%.

(k) AutoAugment

(Cubuk, Zoph, Vasudevan, et al., n.d.) developed a new procedure to automatically search for improved data augmentation policies and termed it as AutoAugment. Autoaugment is a reinforcement learning algorithm that searches for an optimal policy for augmentation. It learns a policy that consists of many sub-policies and each sub policy consists of an image transformation.

(l) Fast AutoAugment

[Lim et al. n.d.] proposed an algorithm to search best augmentation policies in a more efficient way by using searching based on density matching between a pair of train datasets. Fast Autoaugment in comparison to Autoaugment speeds up the search time to find the best policy.

(m) Population Based Augmentation

[Ho et al. n.d.] proposed an algorithm called **population-based augmentation (PBA)** which is helpful in choosing an effective augmentation strategy from a large search space. PBA trains a series of population of neural networks parallelly with some random hyper parameters and optimizes their output to find best optimal state quickly.

(n) RandAugment

(Cubuk, Zoph, Shlens, et al., n.d.) rethinks about the process of designing automated augmentation strategies. As earlier it's not clear that the optimized hyperparameter found for the proxy task is also optimal for the actual task or not. The main concept behind

RandAugment is to improve earlier introduced automated augmentation strategy. To reduce computational expense a simplified search space is proposed to search a single distortion magnitude that jointly controls all operation without any separate search space for proxy task as earlier a search for both magnitude and probability of each operation is done independently for each proxy task, searching for each task consume more time.

(o) Adversarial AutoAugment

(X. Zhang et al., n.d.) proposed an adversarial method to automate augmentation and termed it as Adversarial AutoAugment. The proposed method tries to increase the training loss of the target network by generating adversarial policies for augmentation so that the target network can learn more robust features to improve generalization.

7 COMPARATIVE ANALYSIS

In Section 6, various augmentation methods are discussed starting from the human heuristic approach to the meta-learning approach. All these methods yield significant results and also improve the performance of the training model. In this section, a comparative analysis between meta-learning techniques, GAN, and neural style transfer approaches to augment data will be discussed. The future scope of augmentation relies on these techniques as they can be applied to different domains and have made incredible progress in various real-life applications. Table 5 represents comparative analysis of previously proposed augmentation techniques by researchers with their following details as shown in table below.

Datasets Augmented

Augmentation methods discussed in the previous section perform remarkably well in augmenting some benchmark datasets. In this section, benchmark datasets are listed in a tabular format with the list of models used and augmentation techniques. The table represents test accuracy of various augmentation techniques on some benchmark datasets.

CIFAR-10 (Canadian Institute for Advanced Research) is a collection of images used to train deep learning models. It contains 60,000 colour images (32×32) in 10 different classes. These 10 different classes represent ships, frogs, trucks, horses, deer, dogs, birds, cars, cats, and airplanes. CIFAR-10 images are augmented by different augmentation techniques and test accuracy results are noted as represented by Table 6 below:

Table 6. Performance Analysis of Augmented CIFAR-10 Dataset

Sr. No.	Model	Baseline	Cutout	AA	Fast AA	PBA	Adv. AA
1	Wide-ResNet-28-10	96.1	96.9	97.3	97.3	97.4	98.1
2	Shake-Shake (26 $2 \times 32d$)	96.4	96.9	97.5	97.5	97.4	97.6
3	Shake-Shake (26 $2 \times 96d$)	97.1	97.4	98	98	97.9	98.1
4	Shake-Shake (26 $2 \times 112d$)	97.1	97.4	98.1	98.1	97.9	98.2
5	PyramidNet + Shake Drop	97.3	97.6	98.5	98.3	98.5	98.6

Figure 6 represents the test accuracy value analysed after applying different augmentation techniques on CIFAR-10 dataset.

CIFAR 100 dataset is similar to CIFAR-10 dataset. It consists of 100 classes which contain 600 images each. CIFAR-100 images are augmented by different augmentation techniques and test accuracy results are noted as represented by Table 7 below:

Table 7. Performance Analysis of Augmented CIFAR-100 Dataset

Sr. No.	Model	Baseline	Cutout	AA	Fast AA	PBA	Adv. AA
1	Wide-ResNet-40-2	75.4	74.8	78.5	79.4	-	-
2	Wide-ResNet-28-10	81.2	81.6	82.9	82.7	83.3	84.5
3	Shake-Shake (26 2×96d)	82.9	84	85.7	85.4	84.7	85.9
4	PyramidNet + Shake Drop	86	87.8	89.3	88.3	89.1	89.6

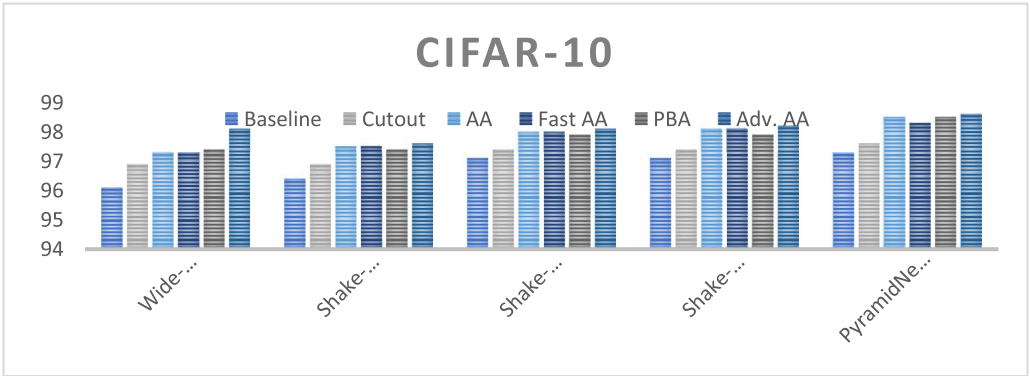


Fig. 6. Performance analysis of Augmented CIFAR-10 dataset.

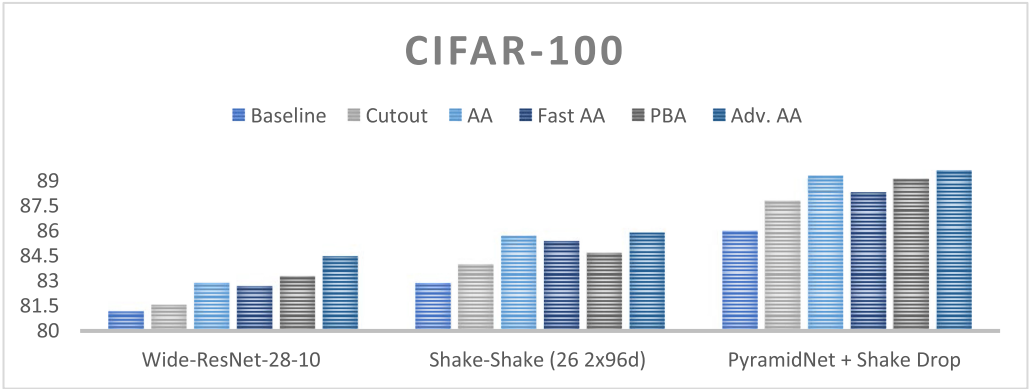


Fig. 7. Performance analysis of Augmented CIFAR-100 dataset.

Figure 7 represents the test accuracy value analysed after applying different augmentation techniques on CIFAR-100 dataset.

SVHN (Street View House Numbers) Dataset is real world images of house numbers obtained from Google Street View images. It contains 73,257 digits for training, 26,032 digits for testing, and 531,131 additional images. SVHN images are augmented by different augmentation techniques and test accuracy results are noted as represented by Table 8 below.

Figure 8 represents the test accuracy value analysed after applying different augmentation techniques on SVHN dataset.

Table 8. Performance Analysis of Augmented SVHN Dataset

S. No.	Model	Baseline	Cutout	AA	Fast AA	PBA	Adv. AA
1	Wide-ResNet-40-2	98.2	98.4	98.7	-	-	98.7
2	Wide-ResNet-28-10	98.5	98.7	98.9	98.9	98.8	99.0
3	Shake-Shake (26 2×96d)	98.6	98.8	99.0	-	98.9	-

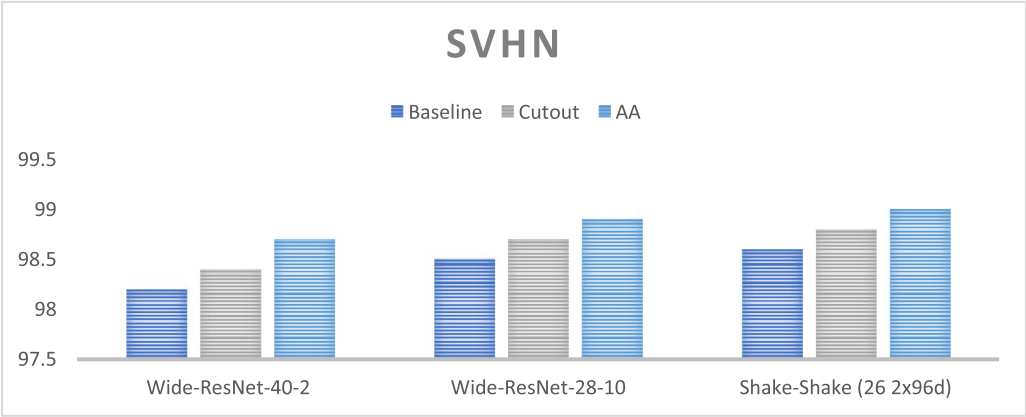


Fig. 8. Performance analysis of Augmented SVHN dataset.

ImageNet is a large image dataset of annotated photographs basically used for research purposes. It contains more than 14 million images in the dataset with more than 21 thousand classes and with one million images that have boundary box annotations. ImageNet images are augmented by different augmentation techniques and test accuracy results are noted as represented by Table 9 below.

Figure 9 represents the test accuracy value analysed after applying different augmentation techniques on ImageNet dataset.

8 APPLICATION AREAS

Recent advancements in augmentation techniques increase its usage in various real-life applications. Figure 10 represents various application domains of data augmentation. Earlier used basic transformation techniques are applicable only on image datasets but with the introduction of GAN network and meta-learning methods to augment data, hence generating synthetic data is possible in different domains. In this section, different application areas are listed where data augmentation improves the performance as represented by Table 10 below.

- (a) **Computer Vision:** Computer vision is a field of artificial intelligence in which models are trained on images and videos to deal with the visual world. It is useful to extract useful information from videos and images and used in various applications which include video surveillance, motion analysis, object detection, etc. Data augmentation is successfully applied in the field of computer vision which improves the performance of various vision tasks. [Du et al. 2020] proposed an object-adaptive LSTM network for the real time visual tracking with adversarial data augmentation. LSTM network fully exploits the sequential dependencies and helps to effectively adapt to the object appearance variation in a complex scenario. They also used a matching based tracking method for the selection

Table 9. Performance Analysis of Augmented ImageNet Dataset

Sr. No.	Model	Baseline	AA	Fast AA	RA	Adv. AA
1	ResNet-50	76.3	77.6	77.6	77.6	79.4
2	ResNet-200	78.5	80.0	80.6	-	81.3
3	EfficientNet-B5	83.2	83.3	-	83.9	-
4	EfficientNet-B7	84	84.4	-	85.0	-

Table 10. Data Augmentation Application in Various Domains

Application Field	Reference	Network	Dataset	Task
Computer Vision	[Du et al. 2020]	LSTM	Public Tracking Dataset: OTB (OTB-2013 and OTB-2015), TC-128, UAV-123 and VOT-2017	Visual Tracking with adversarial data augmentation
	[Songyan Liu et al. 2020]	APGAN	CitySpaces, MPII, Caltech, KITTI, INRIA, ETH and TUD-Brussels	Pedestrian Detection
	[Sultani and Shah 2021]	GAN	UCF-ARG-Aerial, YouTube-Aerial	Human Action Recognition in Drone Videos
Natural Language Processing	[Shakeel et al. 2020]	GAN	Quora, SemEval, MSRP	Paraphrase Detection
	[Qian et al. 2019]	GAN	Aurora4, AMI	Speech Recognition
	[Haralabopoulos et al. 2021]	LSTM	MPST, SEMEVAL, TOXIC, ISEAR, ROBO, AG, CROWD and PEMO	Text Permutation Augmentation
	[Sisi Liu et al. 2020]	Bi-LSTM	BC3, EnronFFP and PA	Sentiment Classification
Security	[Xiang Wang et al. 2020]	GAN	CelebA dataset	Face Augmentation
	[Dhiraj and Jain 2019]	FRCNN	GDX-Ray Dataset	Object Detection in X-Ray images
	[Andresini et al. 2021]	GAN, CNN	CICIDS17, KDDCUP99, UNSW-NB15 and AAGM17	Intrusion Detection
	[Cheng 2019]	CNN GAN	Real Traffic data	Network Traffic Generator
	[P. Wang et al. 2020]	CGAN	ISCX2012, USTC-TFC2016	Encrypted Traffic Classification
Healthcare	[Waheed et al. 2020]	ACGAN	IEEE Covid Chest X-Ray Dataset, Covid-19 Radiography database and Covid-19 Chest X-Ray Dataset Initiative.	Coronavirus Detection
	[Frid-Adar et al. 2018]	GAN	Sample data of Cyst, Metastasis and Hemangioma liver lesions.	Classification of Liver Lesion Problems
	[Chaitanya et al. 2021]	GAN	Cardiac, Prostate, Pancreas	Medical Image Segmentation

of a high-quality dataset to feed it to the LSTM network. To solve the problem of sample inadequacy and class imbalance they used GAN to create augmented data that facilitates the training of the LSTM network. [Songyan Liu et al. 2020] proposed a data augmentation approach for pedestrian detection. This approach helps to tackle the problem of insufficient training data coverage by transferring the source pedestrians to a target scene, and

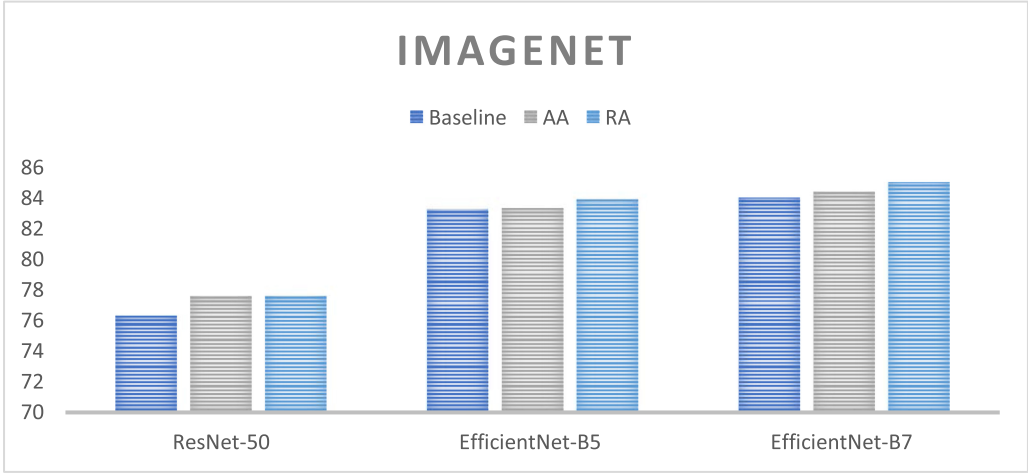


Fig. 9. Performance analysis of Augmented ImageNet dataset.

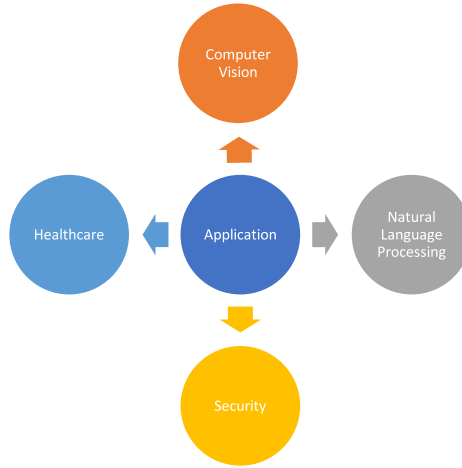


Fig. 10. Data augmentation application in various domains.

then transferring its style by **APGAN (Attribute Preserving Generative Adversarial Networks)**. It helps by providing variation in the dataset and the proposed method yields significant results to improve the generalization ability of the detector and enhance its robustness. [Sultani and Shah 2021] proposed a framework for human action recognition in drone videos, they used YouTube drone videos as dataset and GAN network to augment the video dataset. They demonstrate that features from aerial game video and GAN generated video help in improving action recognition in real aerial videos.

- (b) **Natural Language Processing:** Natural Language Processing is a subfield of artificial intelligence that focus on the interaction between computer and human natural language data i.e., how we can model a system to manage large amounts of natural language data. It includes some tasks i.e., speech recognition, natural language generation, and understanding. Data augmentation is used in various natural language processing tasks and builds an improved model which can generalize efficiently. [Shakeel et al. 2020] proposed

a data augmentation strategy and a multi cascaded model for paraphrase detection. The augmentation method generates paraphrase and non-paraphrase annotations based on graph analysis of existing annotations. The multi cascaded model employs multiple feature learners to encode and classify short text pairs. This approach yields significant improvement results of deep learning models for paraphrase detection. [Qian et al. 2019] proposed a new framework for robust speech recognition using generative adversarial networks. Synthetic data generated frame by frame based on spectrum feature level by using basic GAN. There are no true labels that existed for them and this unsupervised learning framework is used for acoustic modelling. For better data generation conditional GAN is used which explored two different conditions to provide true labels directly. Then during acoustic model training, these true labels are combined with soft labels for improving the performance of the model. [Haralabopoulos et al. 2021] proposed a framework for text permutation augmentation by using sentence permutation to augment an initial dataset. This permutation method improves accuracy by an average of 4.1%. Negation and Antonym augmentation further improve classification accuracy by 0.4% when compared to the permutation augmentation method. [Sisi Liu et al. 2020] develop a framework for document-level multi-topic sentiment classification of E-mail data. Bi-LSTM network is used to model structural dependencies on a topic level within documents and LDA with text segmentation to transfer documents into topic segment. Large volumes of labelled E-mail data are rarely publicly available, so they used data augmentation to create synthetic data for the training model which helps in improving the performance of the model.

- (c) **Security:** Data Augmentation is applicable to various tasks which are helpful for security purposes. Object detection and face recognition can be helpful in an organization to identify a person or an object to secure the system. [Xiang Wang et al. 2020] studied various data augmentation approaches applied for face related tasks. It gives an outline of how to do face augmentation and what it can do. They perform different face data transformations which include geometric and photometric transformation, hairstyle transfer, facial makeup transfer, accessory removal or wearing, pose transformation, expression synthesis and transfer, age progression and regression, and some other types of transformations to enrich the face dataset. All these augmentation methods can be used to improve the robustness of the model by increasing the variation of training data. [Dhiraj and Jain 2019] studied various object detection strategies for threat object detection in baggage security imagery. Baggage screening through X-ray is manually done to recognize the potential threat objects. In the proposed work deep learning framework is used for threat object detection by generating new X-ray images. [Andresini et al. 2021] used GAN based data augmentation for an imbalance dataset of images for classification of network traffic. [Cheng 2019] used (CNN) GAN for generating a network traffic dataset such as ICMP Pings, DNS queries and HTTP web requests. [P. Wang et al. 2020] proposed a traffic data augmenting method by using Conditional GAN which can control modes of data to be generated and termed it as PacketCGAN. It achieves remarkable results in classifying an encrypted traffic dataset.
- (d) **Healthcare:** Deep learning frameworks are widely used in the healthcare domain. Building a deep learning model requires a lot of data but in the medical field, sufficient data is either not available or is not shared because of privacy concerns. In this year of pandemic disease, a lot of work is going on to build deep learning models for pattern recognition, risk estimation of COVID-19 using chest X-rays. A limited amount of data is available to train the model for COVID-19 infected X-ray images. Transfer Learning and Data Augmentation is applied to deal with the problem of data scarcity and improve the performance of

deep learning models. [Waheed et al. 2020] proposed a model (Covid GAN) to generate chest X-ray images by using **Auxiliary Classifier Generative Adversarial Network (ACGAN)** which enhances the performance of CNN for coronavirus detection. GAN is also applicable in various health related issues, it performed remarkably well in the classification of liver lesion problems [Frid-Adar et al. 2018]. [Chaitanya et al. 2021] proposed a semi supervised task-driven data augmentation method for medical image segmentation by using GAN network. Synthetic images help in improving segmentation performance.

9 CONCLUSION

Deep Learning revolutionizes our everyday life because of its successful application in various real-world problems. Data scarcity is one of the major challenges while building a deep learning model as a lot of data is required to train the model so that it can generalize accurately whenever tested for some unseen data. Cost Sensitive Learning, Transfer Knowledge and Data Augmentation can be used to deal with limited data as discussed in Section 5. Data Augmentation has been widely used in various applications and results in improvement of learning of deep learning models by augmenting data of various domain as discussed in Section 8. In Section 6, various data augmenting methods are listed which not only augment image data by basic image manipulation but also augment audio, video, and text data. Recently introduced augmentation techniques increase its application as discussed in Section 8. It is applicable on data of various domains like in Computer Vision, Natural Language Processing, Healthcare, and Security. Data Augmentation can be achieved by various methods and has a great scope for future research as augmentation methods help in improving the performance of the deep learning model by improving generalization and reducing the problem of model overfitting.

10 FUTURE SCOPE

Future work in Data Augmentation will be focused on using meta-learning approaches for augmenting training data. It also focuses on combining meta-learning approaches with other augmentation techniques to improve the performance of the deep learning model. Earlier data augmentation is applicable only on image datasets but with recently introduced techniques its application domain extended to augment text, video, and audio dataset. GAN is applicable to various domains as studied in Section 8. GAN network faces the mode collapse problem, in future work its quality can be improved by solving its mode collapse issue to augment various datasets. A combination of GAN network and meta-learning architecture is an area to be explored by future researchers to build an advanced model. Augmentation tools can be designed to augment data efficiently. Adding more data helps in improving the overall performance of models, and with recently introduced Meta-Learning approaches, Adversarial augmentations, and neural style transfer, it will help researchers overcome the scarcity of data in various domains and improve deep learning models.

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